

1. Use the **limit of Riemann sum definition** the definite integral to evaluate $\int_0^2 (2 - 3x^2) dx$.

Note: You may use right-endpoint Riemann sum. No credit will be given if you use other method.

Formulas you may need: $\sum_{i=1}^n c = cn$, $\sum_{i=1}^n i = \frac{n(n+1)}{2}$, $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} = \frac{2n^3 + 3n^2 + n}{6}$,

$$\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^4 + 2n^3 + n^2}{4}$$

$$\Delta x = \frac{2}{n}, \quad \text{right-endpoints: } x_i^* = i \cdot \frac{2}{n} = \frac{2i}{n}, \quad f(x_i^*) = 2 - 3 \cdot \left(\frac{2i}{n} \right)^2 = 2 - \frac{12i^2}{n^2}$$

$$\begin{aligned} \int_0^2 (2 - 3x^2) dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(2 - \frac{12i^2}{n^2} \right) \frac{2}{n} \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{4}{n} - \frac{24i^2}{n^3} \right) = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \frac{4}{n} \right) - \left(\sum_{i=1}^n \frac{24i^2}{n^3} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{4}{n} \cdot n - \frac{24}{n^3} \cdot \frac{2n^3 + 3n^2 + n^2}{6} \right) \\ &= \lim_{n \rightarrow \infty} \left(4 - 4 \cdot \left(\frac{2n^3}{n^3} + \frac{3n^2}{n^3} + \frac{n^2}{n^3} \right) \right) \\ &= \lim_{n \rightarrow \infty} \left(4 - 4 \left(2 + \frac{3}{n} + \frac{1}{n} \right) \right) = 4 - 8 = -4. \end{aligned}$$

2. Use Part 1 of the Fundamental Theorem of Calculus to find the derivative of $F(x) = \int_0^{\ln x} \sqrt{t^3 + 1} dt$.

$$F'(x) = \sqrt{(\ln x)^3 + 1} \cdot \frac{1}{x}.$$